CSE215
Foundations of Computer Science

Proof by Contradiction

State University of New York, Korea

Prove: There is no greatest integer

- Proof.
  - We use proof by contradiction.
  - Assume there exists a greatest integer.
  - Let G denote the greatest integer. We have
    - (A): for any integer n, n <=G.
  - But G + 1 is an integer satisfying G+1>G. This contradicts with (A)
  - Therefore, there does not exist a greatest integer
- QED.

Proof by contradiction idea

P is true

is the same as ~P leads to contradiction

$\sqrt{2}$ is irrational

- Proof.
  - We use proof by contradiction.
  - Assume sqrt(2) is a rational number, namely:
    - (A) there exists two integers m, n such that sqrt(2)=m/n, and m and n have no common factors.
  - Thus m^2 = 2 n^2. Thus, m^2 is even. Thus m must be even (otherwise m^2 becomes odd).
  - Thus m = 2k for some integer k. Thus, n ^2= 2 k^2. Thus n^2 is even and therefore n must be even.
  - But the fact that m and n are both even contradicts with (A)
  - Therefore sqrt(2) must be irrational.
- QED.

Prove: For any prime number p and natural number n,

If $p|n$**, then** $p \nmid (n+1)$**.**

- Proof.
  - We use proof by contradiction
  - Assume:
    - (A) there exists a prime p and a natural number n, such that p | n and p | (n+1)
  - Since p | n, n = pk for some integer k
  - Since p | (n+1), n+1=pk’ for some integer k’
  - Thus 1 = p (k’ - k). Thus p = 1 which contradicts with the fact p is a prime.
  - Therefore (A) is false
- QED

Summary so far

- To prove P is true, we can prove ~P -> False
- In other words, we assume ~P and try to derive a contradiction

Exercises

Exercise 1

If $a, b \in \mathbb{Z}$, then $a^2 - 4b - 3 \neq 0$.

Exercise 2: Prove the following

Suppose $x \in \mathbb{R}$. If $x^2 + 5x < 0$ then $x < 0$.

Exercise 3: Prove the following

Suppose $x \in \mathbb{R}$. If $x^3 - x > 0$ then $x > -1$.

Exercise 4

Prove: For any two integer a and b,

$$a^2 - 4b \neq 2$$

Summary

- To prove P, we only need to prove ~P -> False
- We start by assuming ~P
- We end by showing a contradiction.